

Final Project Report

Omnichannel Retail Sales and Inventory Analytics Dashboard

Submitted By

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Internship Program

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1. Project Summary

The Omnichannel Retail Sales and Inventory Analytics Dashboard project was developed to analyze retail sales data collected from multiple sales channels and provide actionable business insights. The objective was to transform raw retail transaction data into meaningful visualizations that support data-driven decision-making.

Using Python, SQL, and Power BI, an end-to-end analytics solution was created to monitor sales performance, customer behavior, product performance, geographic sales distribution, and customer loyalty trends.

The dashboard enables stakeholders to track key business metrics and identify opportunities for revenue growth, inventory optimization, and customer retention.

2. Project Objectives

The primary objectives of the project were:

- Analyze overall sales performance across multiple channels.
 - Identify top-performing products and categories.
 - Understand customer purchasing behavior.
 - Evaluate channel-wise revenue contribution.
 - Analyze city-wise sales performance.
 - Generate actionable business recommendations.
 - Develop an interactive Power BI dashboard for business users.
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3. Tools and Technologies Used

Technology	Purpose
Python (Pandas)	Data Cleaning and Transformation
SQL	Data Analysis and Aggregation
Power BI	Dashboard Development
Git & GitHub	Version Control
Jupyter Notebook	Data Exploration

4. Data Preparation

The dataset was cleaned and transformed before analysis.

The following preprocessing steps were performed:

- Removal of duplicate records.
- Handling missing values.
- Standardization of data formats.
- Data validation and quality checks.
- Creation of calculated metrics for reporting.

The cleaned dataset was then used for SQL analysis and Power BI visualization.

5. Dashboard Overview

The Power BI dashboard consists of multiple analytical views including:

KPI Cards

- Total Revenue
- Total Orders
- Total Customers

Sales Analysis

- Monthly Sales Trend
- Sales by Channel
- Sales by Product Category

Product Analysis

- Top 10 Products by Revenue

Customer Analysis

- Revenue by Age Group
- Customer Loyalty Distribution
- Gender Distribution

Geographic Analysis

- City-wise Sales Distribution Map

Interactive Filters

- Year
 - Category
 - Channel
 - Gender
 - City
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6. Key Findings and Insights

6.1 Revenue Performance

The business generated:

- Total Revenue: 224 Million
- Total Orders: 10,000
- Total Customers: 1,000

These metrics indicate strong overall business performance and customer engagement.

6.2 Monthly Sales Trend

Analysis of monthly sales revealed noticeable fluctuations throughout the year.

Key observations:

- Sales remained consistently strong across multiple months.
- Revenue patterns indicate seasonal variations in customer demand.
- Certain months generated significantly higher sales than others.

Business Impact

Understanding seasonal demand helps management optimize inventory planning and marketing activities.

6.3 Channel Performance

The dashboard analyzed revenue generated from:

- Store
- Online
- Marketplace
- Mobile App

The Store and Online channels contributed a substantial share of overall sales.

Business Impact

Strong omnichannel performance demonstrates the importance of maintaining both physical and digital retail channels.

6.4 Product Performance

Top-performing products generated revenues ranging between approximately 2.4M and 4.3M.

Business Impact

High-performing products should receive priority inventory allocation and promotional support to maximize revenue.

6.5 Category Performance

Category-wise analysis showed:

- Sports category contributed the highest percentage of sales.
- Electronics, Beauty, and Footwear categories also generated significant revenue.

Business Impact

Investment in high-performing categories can improve profitability and customer acquisition.

6.6 Customer Age Group Analysis

Revenue contribution was highest among:

1. Middle Age Customers
2. Adult Customers

Younger and senior customer groups generated comparatively lower revenue.

Business Impact

Marketing campaigns should focus on the most profitable customer segments while developing strategies to engage underrepresented age groups.

6.7 Geographic Analysis

The dashboard identified major sales contributions from key metropolitan cities including:

- Delhi
- Mumbai
- Kolkata
- Hyderabad
- Chennai

Business Impact

These cities represent strong markets and should be prioritized for inventory allocation and localized marketing campaigns.

6.8 Customer Loyalty Analysis

Customer loyalty was distributed across:

- Silver Members
- Gold Members
- Platinum Members

Gold members represented the largest customer segment.

Business Impact

Expanding loyalty rewards can improve customer retention and repeat purchases.

6.9 Gender Analysis

Sales contributions from male and female customers were relatively balanced.

Business Impact

Marketing strategies should remain inclusive while leveraging category-specific preferences.

7. Business Recommendations

Based on the analysis, the following recommendations are proposed:

Recommendation 1: Optimize Inventory Planning

Increase stock levels for top-performing products and categories to avoid stock shortages.

Recommendation 2: Focus on High-Revenue Categories

Allocate additional marketing resources to Sports, Electronics, and Footwear categories.

Recommendation 3: Strengthen Digital Channels

Expand investments in Online and Mobile App platforms to increase customer reach.

Recommendation 4: Improve Customer Retention

Enhance loyalty programs and introduce personalized rewards for repeat customers.

Recommendation 5: Target High-Value Customer Segments

Develop campaigns specifically designed for Middle Age and Adult customer groups.

Recommendation 6: Regional Expansion Strategy

Focus expansion efforts on cities with strong sales performance.

8. Project Outcomes

The project successfully delivered:

- A cleaned and structured retail dataset.

- SQL-based business analysis.
- Interactive Power BI dashboard.
- Customer and product performance insights.
- Strategic business recommendations.

The dashboard provides decision-makers with a comprehensive view of retail operations and supports data-driven planning.

9. Conclusion

The Omnichannel Retail Sales and Inventory Analytics Dashboard successfully transformed retail transaction data into actionable business intelligence. Through data cleaning, SQL analysis, and Power BI visualization, valuable insights were generated regarding sales trends, customer behavior, product performance, and geographic distribution.

The findings can assist management in improving inventory planning, optimizing marketing efforts, increasing customer retention, and driving overall business growth.

This project demonstrates practical skills in Data Analytics, Business Intelligence, Data Visualization, SQL, Python, and Power BI and serves as a complete end-to-end analytics solution.

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